Gemma 2: Improving Open Language Models at a Practical Size

Gemma Team, Google DeepMind (2024), arXiv:2408.00118

Paper summary for the CS-781 capstone

The paper at a glance

Citation

Gemma Team, Google DeepMind (M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, et al.). *Gemma 2: Improving Open Language Models at a Practical Size.* Technical report, arXiv:2408.00118, 2024.

One-sentence summary

Open decoder-only models (2B, 9B, 27B) that combine interleaved local/global attention, grouped-query attention and other refinements. The 2B and 9B models are trained by **knowledge distillation** instead of plain next-token prediction.

Why it matters for our capstone

Our A4 follows the same recipe at small scale: local + global layers, GQA, RoPE, RMSNorm and a gated FFN. Distillation is mandatory in our Phases 2–7.

Where it appears in our slides

Main deck: “A4: efficient decoder” and “Compression: knowledge distillation”.

Roadmap

1. The problem
2. The basic idea: architecture
3. The basic idea: training by distillation
4. Results
5. Assessment and outlook

Small models are under-trained

- Recent small models improved mainly by **training on more tokens**.
- But gains grow only logarithmically with data: the latest small models needed up to 15T tokens to improve by less than 1–2%.
- So small models are still under-trained, and more raw text is an expensive fix.

Question: can we give a small model *more information per training step* instead of more steps?

The answer: distillation

Replace the one-hot next-token target with the **full probability distribution** from a larger teacher model.

The report uses this to *simulate training beyond the available data*: the 2B and 9B students see more than 50× the compute-optimal number of tokens.

Model overview

	2B	9B	27B
d_{model}	2304	3584	4608
Layers	26	42	46
Query heads / KV heads	8 / 4	16 / 8	32 / 16
Head size	256	256	128
FFN (GeGLU) dim	18,432	28,672	73,728
Sliding window / global span	4,096 / 8,192		
Vocabulary	256,128 (tied embeddings)		
Training tokens	2T	8T	13T
Trained by	distillation	distillation	from scratch

Shared with Gemma 1

Decoder-only Transformer, 8,192-token context, RoPE positions, approximated GeGLU, tied embeddings.

Embeddings are costly here

The 256k vocabulary gives the 2B model 590M embedding parameters (2.02B non-embedding). Our 32k vocabulary costs 24.6M of 82M.

Interleaved local and global attention

Layers alternate: **local** sliding-window attention (4,096 tokens), then **global** attention (8,192 tokens), every other layer.

- Local layers are cheap: cost and KV-cache grow with the window, not the full length.
- Global layers keep long-range information flowing.

Ablation: changing the 9B model's window *only at inference*:

Sliding window	4096	2048	1024
Validation perplexity	1.63	1.63	1.64

Reading it

Shrinking the local window 4× at inference barely changes perplexity. The window can be a knob for a small speed gain.

Our A4 at small scale

A4 uses a 256-token window in local layers and full attention in every *third* layer (Gemma 2: every second). Its context is 1,024 tokens.

Logit soft-capping

Keep logits within $(-c, c)$ with a smooth squash:

$$\text{logits} \leftarrow c \cdot \tanh(\text{logits}/c)$$

- $c = 50$ for attention scores in every layer.
- $c = 30$ for the final output logits.

Small values pass almost unchanged; very large values are squeezed toward $\pm c$. This prevents a few extreme logits from destabilising training.

Easy example ($c = 30$)

Raw logit	Capped
5	$30 \tanh(0.167) = 4.95$
10	$30 \tanh(0.333) = 9.65$
30	$30 \tanh(1) = 22.85$
60	$30 \tanh(2) = 28.9$
300	≈ 30.0

Ordinary logits barely move; a runaway logit of 300 is held at 30.

Normalisation and grouped-query attention

Pre-norm *and* post-norm with RMSNorm

RMSNorm is applied to both the **input and the output** of each sub-layer (attention and FFN), for training stability:

$$x \leftarrow x + \text{RMSNorm}\big(f(\text{RMSNorm}(x))\big)$$

Our A4 uses only pre-norm.

Grouped-query attention

Two query heads share each KV head (`num_groups = 2`).

Ablation (9B), average over 4 benchmarks:

MHA	50.3
GQA	50.8

Little change in quality, with fewer parameters and faster inference.

Knowledge distillation as the pre-training objective

For each position, the teacher gives a distribution $P_T(x \mid x_c)$ over the next token. The student minimises the cross-entropy to it:

$$\min_{P_S} \sum_x -P_T(x \mid x_c) \log P_S(x \mid x_c)$$

- Standard training uses a one-hot target: only the true next token counts.
- Distillation gives a *richer gradient*: every plausible token gets credit.
- Used for the 2B and 9B models. The 27B model was trained from scratch.

Easy legal example

Context: "The appeal is hereby ___". True next word: "dismissed".

One-hot target: dismissed 1.0.
Teacher target: dismissed 0.62, rejected 0.20, allowed 0.10, disposed 0.05, ...

From one example the student learns four useful facts instead of one, much as in Hinton et al. (2015).

Does distillation really help? Ablations

2B model, 500B tokens, from scratch vs distilled from a 7B teacher:

	Average (3 benchmarks)
From scratch	60.3
Distilled	**67.7**

Across student sizes (validation perplexity, 7B teacher):

	200M	400M	1B
From scratch	23	19	17
Distilled	**21**	**17**	**15**

Reading it

- A 7.4-point average gain at the same data and size.
- The gain holds as the student grows from 200M to 1B.
- 200M is close to our size range (50–300M), so the result is directly relevant to us.

Wide vs deep (9B)

At equal parameters, a **deeper** network scored 52.0 vs 50.8 for a wider one: a small but consistent gain.

Pre-trained model results

Benchmark	Gemma-1 2B	Gemma-2 2B	Gemma-1 7B	Gemma-2 9B	Gemma-2 27B
MMLU	42.3	52.2	64.4	71.3	75.2
GSM8K	15.1	24.3	51.8	68.6	74.0
ARC-C	48.5	55.7	61.1	68.4	71.4
TriviaQA	53.2	60.4	63.4	76.6	83.7
HumanEval	22.0	20.1	32.3	40.2	51.8
Average (8)	44.0	50.0	62.4	70.2	74.4

For comparison, Mistral 7B averages 61.0 and LLaMA-3 8B 61.9 on the same 8 benchmarks.

Reading it

Gemma-2 9B beats LLaMA-3 8B by about 8 points on average. Gemma-2 2B beats Gemma-1 2B while trained on *fewer* tokens (2T vs 3T); the report attributes the gain mainly to distillation, though the architecture changed too. The report claims the models compete with models 2–3× larger.

Post-training, robustness and safety

Instruction tuning:

- supervised fine-tuning on synthetic and human prompts, with responses mostly written by a larger *teacher*;
- on-policy distillation from the teacher;
- RLHF with a reward model much larger than the policy;
- **model merging**: averaging models trained with different hyperparameters.

Format robustness

Standard deviation of MMLU over 12 prompt and evaluation formats:

Gemma 2 2B / 9B / 27B	2.1 / 0.9 / 1.0
Mistral 7B	6.9

Responsible release

Data filtering for personal and unsafe content, evaluation-set decontamination, and safety evaluations. Users are told to run their own safety tests before deployment.

Lessons for our capstone

Gemma 2 finding	What it suggests for us
Distillation beats from-scratch for small students	Distil A4-small from A1 in Phase 1; expect gains in the mandatory Phase 2–7 distillation too
Gain holds from 200M to 1B	Our 47–300M range is in scope
Local window can shrink at inference	Try A4 with windows of 128/256/512 at generation time
GQA $\approx$ MHA in quality	A4's 4 KV heads should cost little quality
Deeper beats wider	At fixed parameters, test more layers with a smaller d_{model}
Soft-capping stabilises training	A cheap addition if fp16 training on a T4 diverges
Model merging helps	Average checkpoints from different runs before submission

Strengths and limitations

Strengths

- Strong open models at practical sizes.
- Clear ablations for distillation, GQA, depth and window size.
- Shows distillation as a way to go beyond available data.
- Combines several proven techniques in one design.
- Open weights for the community.

Limitations

- A technical report: training data and many details are not public.
- Ablations use few benchmarks, often averages over 3–4.
- Needs a strong, expensive teacher, and teacher inference over trillions of tokens.
- Mostly English; not built for strong multilingual use.
- The authors list factuality, adversarial robustness, reasoning and alignment as open problems.

Future research directions

Stated by the authors

- Improve **factuality**.
- Improve **robustness to adversarial attacks**.
- Improve **reasoning**.
- Improve **alignment**.

Natural follow-ups

- How small can the teacher–student gap be before distillation stops paying off?
- Cheaper distillation: store only the teacher's top-k probabilities.

Open for legal NLP

- A law firm could distil a large in-house model into a laptop-sized one, keeping client data private: the motivation of this capstone.
- Local attention suits long judgments, but party names and statutes from page 1 must reach page 40. How many global layers are enough?
- Can distillation from a general teacher transfer legal knowledge, or must the teacher itself be legal?

Summary

- Small models are under-trained, and more raw data gives only logarithmic gains.
- **Gemma 2**: decoder-only, with interleaved local (4,096) and global (8,192) attention, GQA, RoPE, GeGLU, pre+post RMSNorm and logit soft-capping.
- The 2B and 9B models are trained by **distillation** from a larger teacher, over 50$\times$ the compute-optimal token count.
- Ablations: distillation 67.7 vs 60.3 from scratch; gains hold from 200M to 1B; GQA $\approx$ MHA; deeper beats wider; the window can shrink at inference.
- Gemma-2 9B averages 70.2 vs 61.9 for LLaMA-3 8B.
- For us: a large-scale version of A4's design, and strong evidence for our distillation experiments.

Reference

Gemma Team, Google DeepMind (2024). Gemma 2: Improving Open Language Models at a Practical Size. arXiv:2408.00118.

Other work cited: Hinton et al. (2015) Distillation; Beltagy et al. (2020) Longformer (local attention); Shazeer (2020) GLU Variants; Su et al. (2021) RoPE; Hoffmann et al. (2022) Compute-optimal training; Ainslie et al. (2023) GQA; Agarwal et al. (2024) and Gu et al. (2024) on-policy distillation; Ramé et al. (2024) model merging.